

From 2D Graph-Convolutional Newton Methods to Peripheral Vascular Impedance Imaging

Original Injun Lee implementation, implemented nonlinear PVI adaptation, mathematical derivation, experiments, results, and validation roadmap

Technical project report

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Abstract

Electrical impedance tomography (EIT) seeks an interior conductivity field from boundary-current and boundary-voltage measurements. The problem is nonlinear, ill-conditioned, and severely underdetermined. Injun Lee’s original two-dimensional Graph-Convolutional Newton Method (GCNM) combines a Levenberg–Marquardt (LM) update with a graph convolutional network (GCN) on the finite-element mesh. At every learned stage, the original code recomputes the nonlinear forward solution and Jacobian at the current conductivity estimate; a new stage-specific GCN then maps the current estimate and LM update directly to the clean conductivity target.

This report derives that method from the complete electrode model, audits the original code step by step, and compares it with our adaptation for an eight-electrode peripheral vascular impedance (PVI) ring. Our PVI system has only 32 measurements per frame, compared with 928 in the original 32-electrode PyEIT protocol. We established production-pipeline parity for subject 6 and the US120 ring, constructed matched 5,320-element forward and 1,330-element inverse meshes, and trained on clean synthetic vascular conductivity changes rather than PVI images. The earlier fixed-feature experiment improved nonlinear-holdout image correlation from 0.265 (Newton) to 0.463, but increased background energy and image root-mean-square error (RMSE). We therefore implemented the defining GCNM mechanism: every learned stage now recomputes the PVI forward solution, nonlinear residual, Jacobian, and LM proposal. On 32 exact nonlinear fine-mesh holdout phantoms, the selected deployable homogeneous-baseline model reaches image correlation 0.557, image RMSE 0.00432, Dice 0.485, and background RMS 0.00128; the matched two-step LM-only control reaches correlation 0.291. Real subject-6 outputs are evaluated separately against archived PVI pseudo-reconstructions because paired anatomical conductivity truth is absent.

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1 Purpose, audience, and principal conclusion

This document is written for two audiences simultaneously. A telecommunications graduate should be able to follow the signal model, inverse problem, and algorithmic flow without prior EIT specialization. A doctoral reader should find explicit model assumptions, weak forms, normal equations, identifiability arguments, implementation distinctions, experimental controls, and unresolved research risks.

Engineering interpretation. GCNM is not a neural network that directly converts voltages into an image. It is a learned iterative inverse solver. Physics proposes a Newton step, and a graph network decides how that step should be spatially regularized on the FEM mesh.

The central audit result is:

$$\boxed{\text{Original GCNM: new } F(\boldsymbol{\sigma}_k), J_k, \delta\boldsymbol{\sigma}_k \text{ at every stage}} \quad (1)$$

whereas the earlier clean-truth PVI experiment used

$$\boxed{\text{Earlier experiment: one fixed Newton feature reused for two stages.}} \quad (2)$$

The implemented faithful adaptation now satisfies

$$\boxed{\text{Implemented PVI-GCNM: new } F, J, \boldsymbol{r}, \boldsymbol{p} \text{ at every stage.}} \quad (3)$$

This lets the report preserve the earlier residual graph denoiser as project history while evaluating the nonlinear unrolled solver as a separate method.

2 System and notation

2.1 Physical system

Eight electrodes are distributed around a wearable ring surrounding a limb cross section. Current is injected through electrode pairs and differential voltages are measured on the boundary. The reconstruction target is conductivity on triangular finite elements and, for visualization, a 40×40 mapped image.

Table 1: Primary dimensions of the original and PVI systems.

Quantity	Injun original	Subject-6 PVI
Physical electrodes	32	8
Stimulation patterns	32	8
Measurements per stimulation	29	4
Measurements per frame M	928	32
Training samples in first full experiment	500	512
Validation samples	internal 20% split	128 independent
Linearized test samples	100 in original script	128
Exact nonlinear test samples	not separated	32
Inverse elements K	mesh-dependent	1,330
Forward elements	mesh-dependent	5,320
Display image	FEM plot	40×40

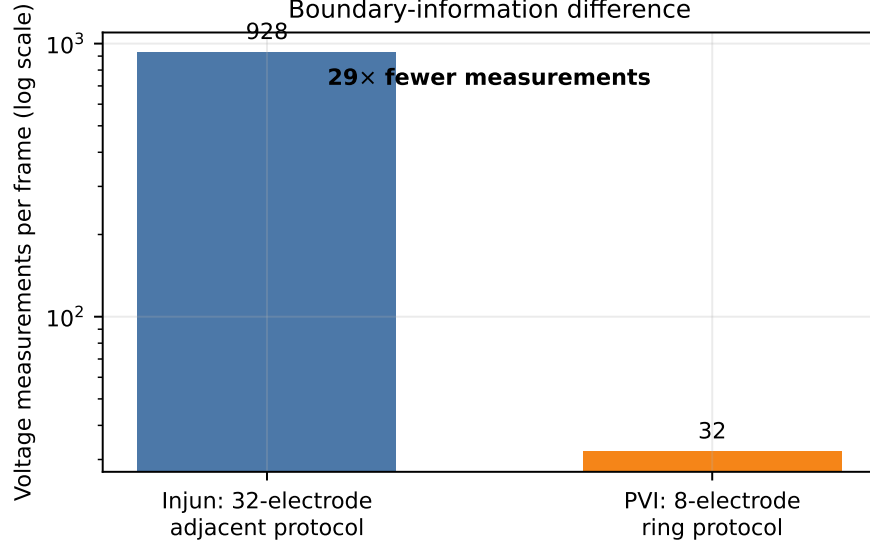


Figure 1: The original adjacent 32-electrode protocol supplies 928 scalar voltage measurements per frame, whereas the PVI ring supplies 32. This information deficit is fundamental and cannot be removed by network capacity.

2.2 Notation

Table 2: Notation used throughout the report.

Symbol	Meaning
$\Omega \subset \mathbb{R}^2$	Cross-sectional conductive domain.
$\sigma(\mathbf{x})$	Conductivity field, in siemens per metre.
$u(\mathbf{x})$	Interior electric potential.
e_ℓ	Boundary surface of electrode ℓ .
I_ℓ, U_ℓ, z_ℓ	Electrode current, voltage, and contact impedance.
$\boldsymbol{\sigma} \in \mathbb{R}^K$	Elementwise FEM conductivity vector.
$\mathbf{v} = F(\boldsymbol{\sigma}) \in \mathbb{R}^M$	Nonlinear forward map from conductivity to measurements.
$J(\boldsymbol{\sigma}) \in \mathbb{R}^{M \times K}$	Jacobian $\partial F / \partial \boldsymbol{\sigma}$.
R	Spatial regularization operator or projected Laplacian.
$G = (\mathcal{V}, \mathcal{E})$	Graph whose nodes are FEM elements.
$\boldsymbol{\sigma}_k, \delta \boldsymbol{\sigma}_k$	Conductivity estimate and LM update at stage k .
$\boldsymbol{\sigma}^*$	Clean conductivity ground truth.
$\boldsymbol{\sigma}_b, \Delta \boldsymbol{\sigma}$	Baseline conductivity and time-varying conductivity change.

3 From telecommunications field theory to EIT

3.1 Quasi-static derivation

At PVI frequencies and length scales, electromagnetic wave propagation is negligible compared with conduction; the electric field is well approximated as curl-free:

$$\nabla \times \mathbf{E} = \mathbf{0}, \quad \mathbf{E} = -\nabla u. \quad (4)$$

Ohm's constitutive relation is

$$\mathbf{J} = \sigma \mathbf{E} = -\sigma \nabla u. \quad (5)$$

Charge conservation in the absence of interior sources gives

$$\nabla \cdot \mathbf{J} = 0. \quad (6)$$

Combining the three equations yields the EIT field equation

$$\boxed{\nabla \cdot (\sigma \nabla u) = 0 \quad \text{in } \Omega.} \quad (7)$$

Engineering interpretation. The voltage distribution is not transmitted through the tissue as an independent signal channel. It is the solution of a boundary-value problem whose coefficients are the unknown conductivity. Changing conductivity changes all measured boundary voltages in a spatially distributed way.

3.2 Complete electrode model

Point-electrode models neglect contact impedance and finite electrode area. The PVI solver uses the complete electrode model (CEM). For electrode e_ℓ ,

$$u + z_\ell \sigma \frac{\partial u}{\partial n} = U_\ell \quad \text{on } e_\ell, \quad (8a)$$

$$\int_{e_\ell} \sigma \frac{\partial u}{\partial n} ds = I_\ell, \quad (8b)$$

$$\sigma \frac{\partial u}{\partial n} = 0 \quad \text{outside the electrodes}, \quad (8c)$$

$$\sum_{\ell=1}^L I_\ell = 0, \quad \sum_{\ell=1}^L U_\ell = 0. \quad (8d)$$

The last voltage constraint fixes the arbitrary potential reference.

3.3 Weak form and finite-element system

Multiplying [equation \(7\)](#) by a test function w , integrating by parts, and including the CEM terms produces the bilinear form

$$\begin{aligned} a_\sigma((u, \mathbf{U}), (w, \mathbf{W})) &= \int_{\Omega} \sigma \nabla u \cdot \nabla w \, d\mathbf{x} \\ &+ \sum_{\ell=1}^L \frac{1}{z_\ell} \int_{e_\ell} (u - U_\ell)(w - W_\ell) \, ds, \end{aligned} \quad (9)$$

with right-hand side $\sum_{\ell} I_\ell W_\ell$. Expanding u in FEM basis functions converts the problem into

$$A(\sigma)\phi = \mathbf{b}, \quad \mathbf{v} = M\phi, \quad (10)$$

where $A(\sigma)$ is the conductivity-dependent stiffness/CEM matrix, ϕ contains nodal and electrode potentials, and M implements differential measurements.

4 The inverse problem

4.1 Measurement model

The measured vector is

$$\mathbf{y} = F(\boldsymbol{\sigma}^*) + \boldsymbol{\varepsilon}, \quad \boldsymbol{\varepsilon} \sim \mathcal{N}(\mathbf{0}, \Sigma_\varepsilon) \quad (11)$$

in the ideal Gaussian model. A regularized estimate solves

$$\hat{\boldsymbol{\sigma}} = \arg \min_{\boldsymbol{\sigma}} \frac{1}{2} \left\| \Sigma_\varepsilon^{-1/2} (F(\boldsymbol{\sigma}) - \mathbf{y}) \right\|_2^2 + \lambda^2 \mathcal{R}(\boldsymbol{\sigma}). \quad (12)$$

For a quadratic spatial prior,

$$\mathcal{R}(\boldsymbol{\sigma}) = \frac{1}{2} \|R(\boldsymbol{\sigma} - \boldsymbol{\sigma}_{\text{ref}})\|_2^2. \quad (13)$$

From a Bayesian perspective, the first term is the negative log likelihood and the regularizer is the negative log prior. The minimizer is a maximum a posteriori estimate under those assumptions [3, 2].

4.2 Why the PVI inverse is severely underdetermined

The local linearized system has $M = 32$ measurements and $K = 1330$ element values:

$$J \in \mathbb{R}^{32 \times 1330}, \quad \text{rank}(J) \leq 32. \quad (14)$$

Consequently,

$$\dim \ker(J) \geq 1330 - 32 = 1298. \quad (15)$$

At least 1,298 independent element directions are locally invisible to the data. Regularization or learned priors select one solution among many data-consistent fields; they do not create new physical information.

Engineering interpretation. A clear-looking image is not automatically a correct image. With only 32 measurements, a network can learn plausible vessel shapes that the voltages do not uniquely support. Independent nonlinear simulation, physical phantoms, and uncertainty analysis are therefore mandatory.

5 Newton and Levenberg–Marquardt reconstruction

5.1 Linearization

Around $\boldsymbol{\sigma}_k$, use the first-order Taylor approximation

$$F(\boldsymbol{\sigma}_k + \mathbf{p}) \approx F(\boldsymbol{\sigma}_k) + J_k \mathbf{p}, \quad J_k = J(\boldsymbol{\sigma}_k). \quad (16)$$

Let $\mathbf{r}_k = F(\boldsymbol{\sigma}_k) - \mathbf{y}$ and $W = \Sigma_\varepsilon^{-1/2}$. The quadratic subproblem is

$$\min_{\mathbf{p}} \frac{1}{2} \|W(J_k \mathbf{p} + \mathbf{r}_k)\|_2^2 + \frac{\lambda^2}{2} \|R(\boldsymbol{\sigma}_k + \mathbf{p} - \boldsymbol{\sigma}_{\text{ref}})\|_2^2. \quad (17)$$

Differentiating and setting the gradient to zero gives

$$\begin{aligned} & (J_k^\top W^\top W J_k + \lambda^2 R^\top R) \mathbf{p}_k \\ & = -J_k^\top W^\top W \mathbf{r}_k - \lambda^2 R^\top R(\boldsymbol{\sigma}_k - \boldsymbol{\sigma}_{\text{ref}}). \end{aligned} \quad (18)$$

Levenberg–Marquardt often replaces or augments the prior Hessian with λI , stabilizing weakly observed directions [4, 5].

5.2 Original code update

Injun’s code computes a normalized PyEIT Jacobian and uses

$$\delta\sigma_k = - \left(J_k^\top J_k + 0.1I \right)^{-1} J_k^\top (F(\sigma_k) - \mathbf{y}). \quad (19)$$

The update is not simply added to the current conductivity. Instead, $[\sigma_k, \delta\sigma_k]$ becomes the GCN input.

5.3 Production PVI one-step inverse

The strict MATLAB PVI path computes an unweighted coarse-mesh Jacobian J_0 at $\sigma_0 = 0.71$ and the projected fine-mesh Laplacian

$$R_{\text{rec}} = \frac{1}{2} C_{2F}^\top R_{\text{fwd}} C_{2F}. \quad (20)$$

Its fixed inverse matrix is

$$D_1 = (J_0^\top J_0 + \lambda^2 R_{\text{rec}})^{-1} J_0^\top, \quad \lambda = 5 \times 10^{-4}. \quad (21)$$

After reference subtraction, the historical displayed PVI convention is

$$\Delta\sigma_{\text{PVI}} = -D_1(\mathbf{v}_t - \mathbf{v}_0). \quad (22)$$

Finite-difference tests show that the Python FEM Jacobian is the physical derivative $\partial\mathbf{v}/\partial\sigma$. For clean physical conductivity supervision we therefore use the sign-correct feature

$$\Delta\sigma_{\text{Newton}} = +D_1\Delta\mathbf{v}. \quad (23)$$

6 Graph convolution as a learned spatial prior

Let A_G be the element-adjacency matrix. With self-loops, $\tilde{A} = A_G + I$ and degree matrix $\tilde{D}$, a standard GCN layer is

$$H^{(\ell+1)} = \text{phi} \left(\tilde{D}^{-1/2} \tilde{A} \tilde{D}^{-1/2} H^{(\ell)} W^{(\ell)} + \mathbf{b}^{(\ell)} \right). \quad (24)$$

Neighbor aggregation acts like a learned, feature-dependent graph filter [6, 7]. Unlike a fixed Laplacian penalty, nonlinear layers can learn to preserve inclusion boundaries, reject isolated noise, and use patterns that recur in the training distribution.

In an unrolled-optimization interpretation, the GCN approximates a learned regularization or proximal map:

$$\sigma_{k+1} = \mathcal{G}_{\theta_k}(\sigma_k, \delta\sigma_k; G). \quad (25)$$

Each stage has different parameters because the error distribution evolves with k .

7 Injun Lee’s original implementation: step by step

The source audited here is `GCNM_training.py` and `GCNM_testing.py`. The workspace contains code but no saved original-run metrics; this report records the experiment as specified and does not fabricate numerical results.

7.1 Original data experiment

- O1.** Create 32-electrode circle or thorax forward/inverse PyEIT meshes.
- O2.** Create the adjacent protocol: 32 excitations and 29 retained voltage measurements per excitation, or 928 total.
- O3.** Draw 500 phantoms with one to four non-overlapping ellipses.
- O4.** Draw background conductivity in $[0.40, 0.43]$, low inclusions in $[0.15, 0.25]$, and high inclusions in $[0.65, 0.95]$.
- O5.** Rasterize identical ellipse geometry onto fine and coarse meshes.
- O6.** Simulate fine-mesh voltage and add Gaussian noise with standard scale 0.005 mean $|V|$.
- O7.** Construct a graph in which inverse elements sharing any vertex are adjacent.

7.2 Original greedy GCNM training

Initialize $\sigma_0 = \mathbf{1}$. For $k = 0, \dots, 9$:

- G1.** For every sample, set the inverse mesh to the current GCN estimate σ_k .
- G2.** Recompute $F(\sigma_k)$ and normalized J_k .
- G3.** Compute the LM feature using [equation \(19\)](#).
- G4.** Set two node features $H_k = [\sigma_k, \delta\sigma_k]$.
- G5.** Initialize a new GCN with widths $2 \rightarrow 250 \rightarrow 250 \rightarrow 250 \rightarrow 1$.
- G6.** Train the GCN to predict the complete absolute σ^* using plain MSE.
- G7.** Use an 80/20 ordered split, Adam at 0.002, batch size 10, up to 10,000 epochs, and early-stopping patience 200.
- G8.** Apply the best model to all samples; its direct output becomes σ_{k+1} .
- G9.** Save a separate checkpoint for stage k .

The essential point is that the GCN does not output a step length or correction; it directly predicts the next full conductivity. The physics update is a feature.

7.3 Original testing

The test script creates 100 new synthetic samples from the same distribution. Starting from $\sigma_0 = \mathbf{1}$, it recomputes the nonlinear LM update, loads GCN stage k , predicts σ_{k+1} , and repeats. It reports relative conductivity ℓ_1 error, conductivity MSE, and forward-voltage relative ℓ_2 error. The root test script does not include a complete iterative-LM-only control, although our modular extracted port later added one.

Three-repository algorithm flow: original learned solver, production PVI, and faithful adaptation

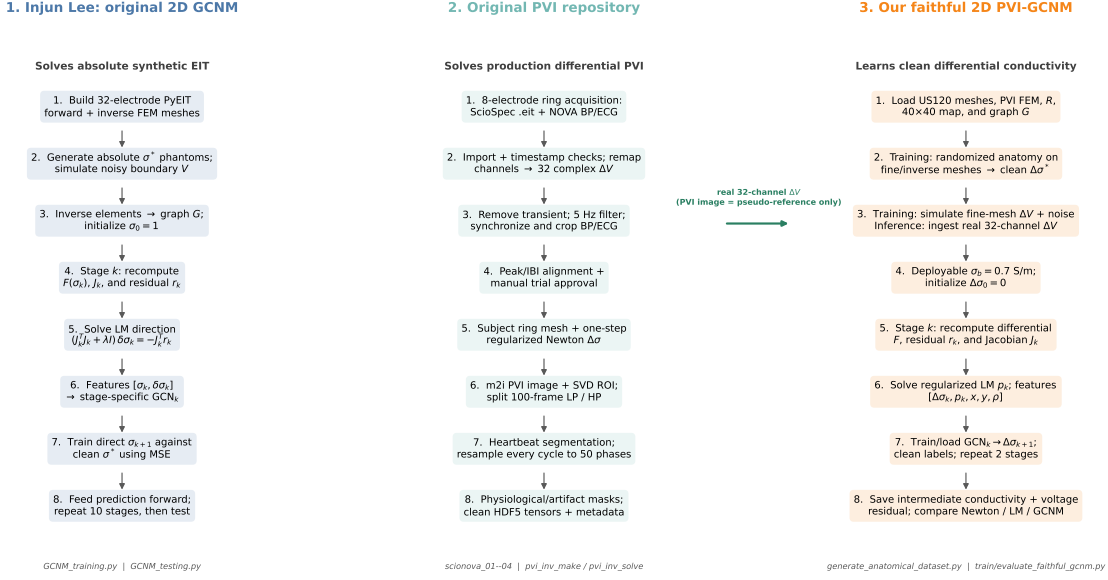


Figure 2: Step-by-step flow across the three repositories. Left: Injun Lee’s original 32-electrode absolute-EIT GCNM. Center: the eight-electrode production PVI acquisition, reconstruction, cardiac segmentation, and export pipeline. Right: our faithful differential PVI-GCNM, which combines clean simulated conductivity supervision with newly recomputed PVI physics at every learned stage. The green cross-lane arrow is important: the PVI repository supplies real 32-channel voltage differences to our inference path; its Newton image is retained only as a pseudo-reference, not as a clean label.

8 Our PVI adaptation

8.1 Production-parity foundation

Before training a learned inverse, we reproduced the subject-6 production chain: raw ScioSpec extraction, electrode remapping, signed 10-mA current, 5-Hz filtering, resistance/reactance conversion, centered-shrink moving average, US120 geometry, projected Laplacian, and MATLAB pixel ordering. This separates two questions:

1. Is the input and deterministic inverse reproduced correctly?
2. Does a learned method improve reconstruction relative to that baseline?

8.2 Clean anatomical targets

PVI images are not scientific labels in the anatomical experiment. We generate skin, fat, muscle, bone, blood, and one or two pulsatile vessels. A shared parameterization is rasterized independently on the 5,320-element forward mesh and 1,330-element inverse mesh. The target is clean vascular $\Delta \sigma^*$.

The training generator has two modes:

$$\text{linearized: } \Delta \mathbf{v} \approx J_{\text{fine},q} \Delta \sigma_{\text{fine}}^*, \quad (26)$$

$$\text{nonlinear: } \Delta \mathbf{v} = F_{\text{fine}}(\sigma_b + \Delta \sigma^*) - F_{\text{fine}}(\sigma_b). \quad (27)$$

The linearized mode uses a bank of randomized anatomical fine-mesh Jacobians for efficient training. The exact nonlinear mode performs two separate FEM solves and is reserved for an independent holdout.

Domain randomization includes static contact impedance, contact drift in nonlinear simulation, channel gain, current gain, white noise, correlated noise, tissue conductivity, layer thickness, bone geometry, vessel position, vessel axes, and cardiac-phase amplitude.

8.3 Earlier fixed-feature residual GCN

The earlier model normalizes conductivity by the 99.5th training percentile s and uses five features:

$$H_i = \left[\frac{\Delta \sigma_{k,i}}{s}, \frac{n_i}{s}, x_i, y_i, r_i \right], \quad n = D_1 \Delta \mathbf{v}. \quad (28)$$

It uses widths $5 \rightarrow 64 \rightarrow 64 \rightarrow 64 \rightarrow 1$. The output is residual around Newton:

$$\widehat{\Delta \sigma}_{k+1} = n + s g_{\theta_k}(H, G). \quad (29)$$

The final graph layer is initialized to zero, so the initial network exactly equals Newton.

Because vessels occupy few elements, the training objective is weighted:

$$w_i = 1 + \alpha \text{clip} \left(\frac{|y_i|}{\text{median}_{j: y_j \neq 0} |y_j|}, 0, 2 \right), \quad (30)$$

$$\mathcal{L} = \frac{1}{SK} \sum_{s=1}^S \sum_{i=1}^K w_{s,i} (\widehat{y}_{s,i} - y_{s,i})^2, \quad \alpha = 8. \quad (31)$$

Two greedy stages were trained, but both reused n . This is the principal departure from the original method.

Table 3: Methodological comparison.

Aspect	Injun original	Earlier fixed-feature PVI experiment
Physics	PyEIT absolute forward problem	PVI CEM differential problem
Target	Absolute σ^*	Vascular $\Delta \sigma^*$
Initial estimate	$\sigma_0 = \mathbf{1}$	$\Delta \sigma_0 = \mathbf{0}$
Update per stage	New $F(\sigma_k), J_k, \delta \sigma_k$	Fixed $n = D_1 \Delta \mathbf{v}$ reused
Features	$[\sigma_k, \delta \sigma_k]$	$[\Delta \sigma_k, n, x, y, r]$
Output	Direct full conductivity	Newton plus learned residual
Stages	10	2
Hidden widths	250, 250, 250	64, 64, 64
Loss	Unweighted MSE	Vessel-weighted MSE
Coordinates	No explicit coordinates	Explicit x, y, r
Data	Generic ellipses	Randomized anatomy and vessels
Main control	Synthetic truth	Newton, linearized and nonlinear truth

8.4 Implemented faithful nonlinear PVI-GCNM

For a fixed inference baseline σ_b and current differential estimate $\Delta \sigma_k$, the implemented stage predicts a differential voltage and forms

$$\widehat{\Delta \mathbf{v}}_k = F(\sigma_b + \Delta \sigma_k) - F(\sigma_b), \quad \mathbf{r}_k = \widehat{\Delta \mathbf{v}}_k - \Delta \mathbf{v}_{\text{meas}}. \quad (32)$$

The PVI solver recomputes $J_k = J(\sigma_b + \Delta\sigma_k)$ and solves

$$\left(J_k^\top J_k + h_{\text{PVI}}^2 R_{\text{rec}} + \lambda_{\text{LM}} I\right) \mathbf{p}_k = -J_k^\top \mathbf{r}_k. \quad (33)$$

Here R_{rec} is the projected inverse-mesh regularizer, h_{PVI} preserves the production inverse scale, and the separate $\lambda_{\text{LM}} I$ stabilizes each stage-dependent solve. The graph input always contains $[\Delta\sigma_k, \mathbf{p}_k]$; the coordinate ablation adds $[x, y, r]$.

Two output parameterizations were executed:

$$\text{direct: } \Delta\sigma_{k+1} = g_{\theta_k}([\Delta\sigma_k, \mathbf{p}_k], G), \quad (34)$$

$$\text{proposal residual: } \Delta\sigma_{k+1} = \Delta\sigma_k + \mathbf{p}_k + c_{\theta_k}([\Delta\sigma_k, \mathbf{p}_k], G). \quad (35)$$

The residual correction layer is zero initialized, and both modes project the absolute conductivity $\sigma_b + \Delta\sigma_{k+1}$ above a positive numerical floor. A separate iterative-LM control applies exactly the same recomputed physics without a GCN. Training and inference bind the baseline policy, physics hyperparameters, mesh, mappings, configuration, and dataset hashes in each checkpoint so an oracle synthetic baseline cannot be silently used on real data.

9 Experiments performed

9.1 E0: original code experiment specification

The original code defines 500 training and 100 testing phantoms, ten greedy GCN stages, and a 250^3 hidden architecture. No original-run result file is present in this workspace. Consequently, the original configuration is used as a method reference, not as a numerical comparator.

9.2 E1: production reconstruction parity

Using 500 frame-matched subject-6 HDF samples, reconstructed high-pass images were compared directly with the archived MATLAB output.

Table 4: Production-parity experiment. This validates the port, not learned image quality.

Metric	Value
Frames	500
Finite pixels	1,364
Finite-mask IoU	1.0000
Global image correlation	0.9936
Median per-frame correlation	0.9914
Minimum per-frame correlation	0.9776
Image RMSE	6.86×10^{-4}
Normalized RMSE	0.1131
Optimal scale	0.9907

9.3 E2: temporal alignment and engineering pseudo-label packs

Automatic segmentation produced 1,310 candidates; 1,298 were monotonically matched to the HDF sequence across 11 trials. Low-pass resistance and reactance mean-channel correlations were 0.99982

and 0.99999. High-pass correlations were lower (approximately 0.67) because archived manual peak merges were unavailable.

A leakage-controlled engineering pack retained 971 clean periods and ten phases per period: 6,930 train samples from trials 1–7, 1,180 validation samples from trials 8–9, and 1,600 test samples from trials 10–11. These labels reproduce PVI and are suitable only for pipeline testing, not superiority claims.

9.4 E3: earlier fixed-feature clean-truth experiment

The full clean experiment used 512 training, 128 validation, and 128 test samples. Each split used eight independently randomized fine-mesh Jacobian backgrounds. Training used two stages, 64^3 hidden widths, batch size 16, learning rate 10^{-3} , 150 requested epochs, patience 15, and $\alpha = 8$ vessel weighting.

9.5 E4: exact nonlinear holdout

Thirty-two independent test phantoms were generated by separate baseline and dynamic fine-mesh FEM solves. They were never used for training or checkpoint selection. This is the stronger distribution-shift test because neither the linearized data approximation nor the training Jacobian bank creates their voltage.

9.6 E5: faithful core and physical control

The faithful core trained two stage-specific GCNs while recomputing [equations \(32\)](#) and [\(33\)](#) at each stage. Controlled pairs used direct versus proposal-residual output, and the same forward/Jacobian code ran two LM iterations with no learning. A saved-anatomy synthetic baseline was retained only as an oracle upper bound. Every deployable experiment trained and inferred with the same homogeneous 0.7 S/m baseline.

9.7 E6: controlled architecture and objective ablations

Starting from the homogeneous direct core, experiments added mesh coordinates, swept vessel weights $\alpha \in \{0, 1, 2, 4, 8\}$, and then swept explicit background penalties $\beta \in \{0.25, 0.5, 1.0\}$ at the selected $\alpha = 1$. The background runs selected checkpoints with a composite normalized-RMSE/background/Dice score. Only one factor changed in each comparison, every experiment used seed 0, and model and result directories were isolated to prevent checkpoint overwrites.

9.8 Compute execution

The production chain, smoke tests, faithful core, and all synthetic ablations completed with exit code 0:0 and empty error logs. Every training and evaluation launcher preserved the requested allocation: one node, one task, one GPU, 16 CPUs, 250 GB, and a one-day wall-time limit on the `ece_bst` partition and account.

Table 5: Executed Slurm experiment groups. Ranges include training and dependent linearized, nonlinear, or real-PVI evaluations as applicable.

Jobs	Purpose	State
426833–426837	Earlier fixed-feature data, training, and evaluation	completed
426840–426844	Faithful end-to-end smoke and voltage-sign verification	completed
426893–426900	Homogeneous direct/residual core and real controls	completed
426901–426915	Coordinates and α sweeps	completed
426921–426929	Background-penalty sweep	completed
426938	Selected-model full real-PVI evaluation	completed

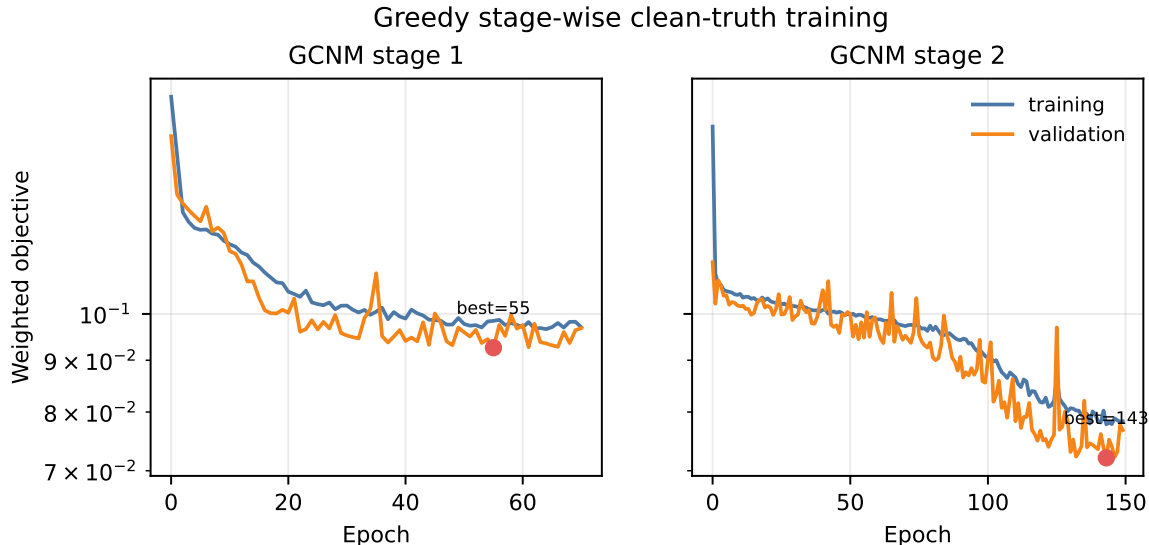


Figure 3: Greedy weighted-objective training. Stage 1 reached its best validation objective around epoch 55; stage 2 continued to approximately epoch 143. Lower weighted loss does not guarantee lower unweighted background RMSE.

10 Quantitative results

10.1 Historical fixed-feature linearized holdout

Table 6: Results on 128 independently generated linearized holdout samples.

Metric	Newton	GCNM stage 1	GCNM stage 2
Element correlation $\uparrow$	0.129	0.329	0.420
Image correlation $\uparrow$	0.209	0.339	0.435
Localization Dice $\uparrow$	0.239	0.357	0.433
Element RMSE $\downarrow$	0.00457	0.00712	0.00674
Image RMSE $\downarrow$	0.00613	0.01134	0.01072
Background RMS $\downarrow$	0.00251	0.00684	0.00653

GCNM finds more correct spatial support, but overestimates structure and background. Stage 2 partially corrects stage-1 amplitude while further improving localization.

10.2 Historical fixed-feature exact nonlinear holdout

Table 7: Results on 32 exact nonlinear fine-mesh holdout samples.

Metric	Newton	GCNM stage 1	GCNM stage 2
Element correlation $\uparrow$	0.175	0.373	0.445
Image correlation $\uparrow$	0.265	0.388	0.463
Localization Dice $\uparrow$	0.295	0.307	0.359
Element RMSE $\downarrow$	0.00354	0.00345	0.00365
Image RMSE $\downarrow$	0.00506	0.00502	0.00541
Background RMS $\downarrow$	0.00063	0.00202	0.00281

Stage 1 is the best balanced estimator: compared with Newton, it more than doubles element correlation, increases image correlation from 0.265 to 0.388, and slightly reduces both element and image RMSE. Stage 2 obtains the best localization and correlation but loses RMSE performance.

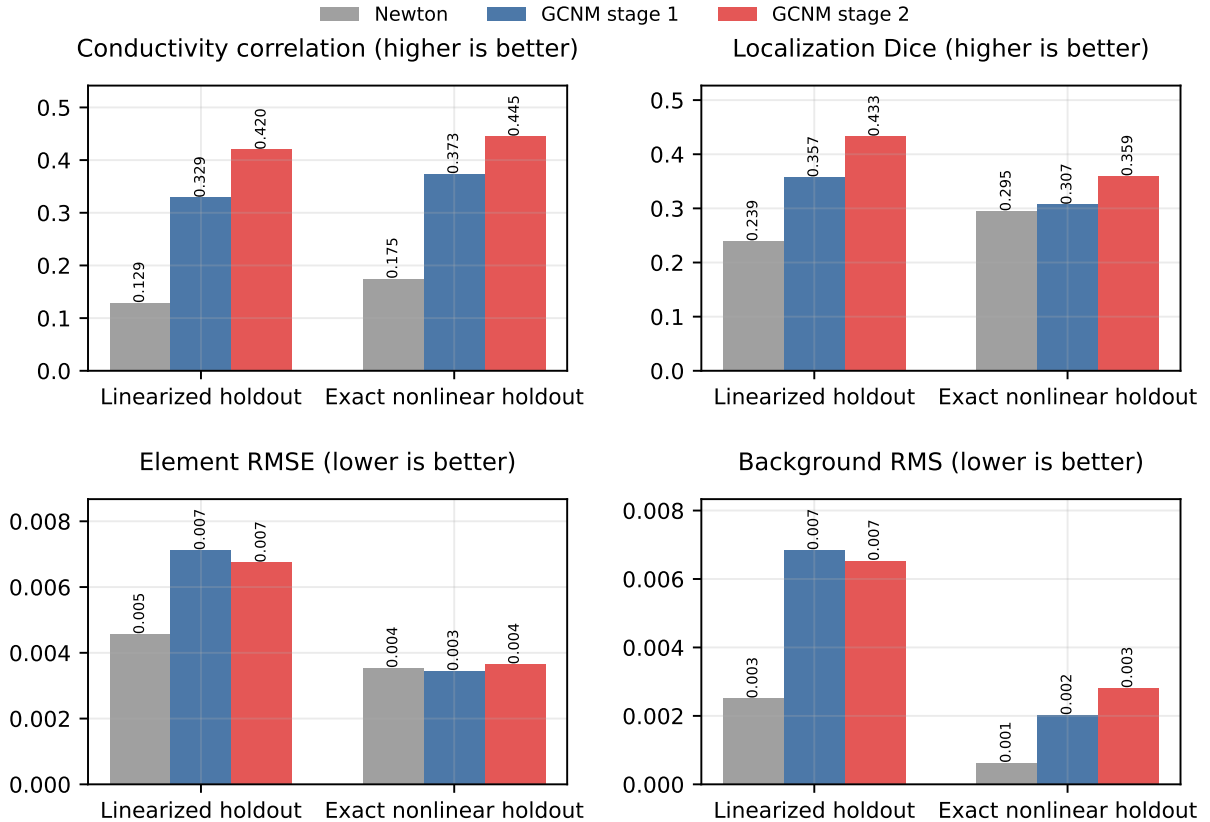


Figure 4: Stage-wise clean-truth metrics. The result is a Pareto trade-off: GCNM improves structure-sensitive metrics but increases false background energy.

10.3 Faithful core and iterative-physics control

The matched core comparison in table 8 uses the same 32 exact nonlinear phantoms. Merely repeating the nonlinear LM solve improves Dice more than the historical saved Newton step but only modestly improves correlation. Learning supplies a larger structural gain. Direct output is stronger than the proposal-residual parameterization on correlation, RMSE, and Dice, so direct output was retained for subsequent ablations.

Table 8: Faithful homogeneous-baseline core on the exact nonlinear holdout. Voltage RMS is the physical residual after stage 2; a dash means it was not recorded for that historical artifact.

Method	Image corr. $\uparrow$	Image RMSE $\downarrow$	Dice $\uparrow$	BG RMS $\downarrow$	Voltage RMS $\downarrow$
Historical saved Newton	0.265	0.00506	0.295	0.000626	—
Iterative LM, stage 2	0.291	0.00503	0.394	0.000589	1.90×10^{-7}
Earlier fixed-feature GCN, stage 2	0.463	0.00541	0.359	0.002810	—
Faithful direct, stage 2	0.458	0.00484	0.230	0.000581	2.73×10^{-6}
Faithful proposal-residual, stage 2	0.390	0.00494	0.176	0.000474	1.36×10^{-6}

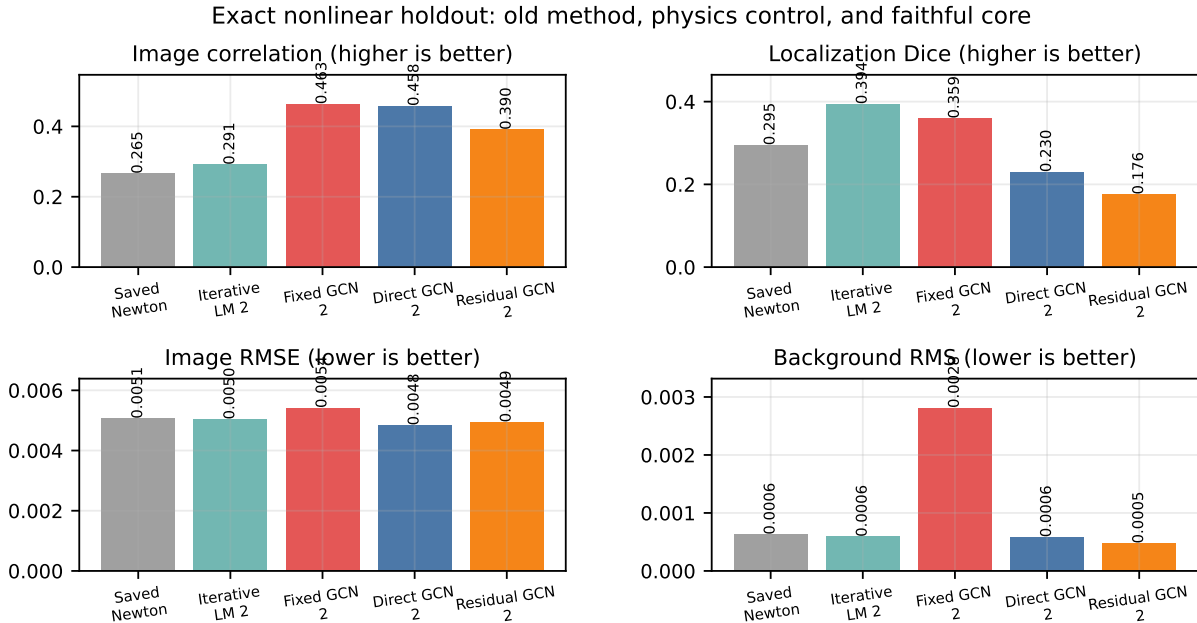


Figure 5: Historical Newton and fixed-feature results, iterative-LM-only control, and the two implemented faithful output parameterizations. No single bar should be read alone: structure, amplitude, localization, and background expose different failure modes.

10.4 Baseline knowledge is an oracle ablation

The saved synthetic σ_b contains static vessel locations. It is therefore information unavailable during deployment, not an acceptable default input. It raises direct-core correlation from 0.458 to 0.520, demonstrating that baseline knowledge matters, but all remaining models use the homogeneous 0.7 S/m policy.

Table 9: Baseline-policy ablation for faithful direct output.

Baseline	Image corr. $\uparrow$	RMSE $\downarrow$	Dice $\uparrow$	BG RMS $\downarrow$
Saved anatomy (oracle only)	0.520	0.00456	0.346	0.000737
Homogeneous 0.7 S/m (deployable)	0.458	0.00484	0.230	0.000581

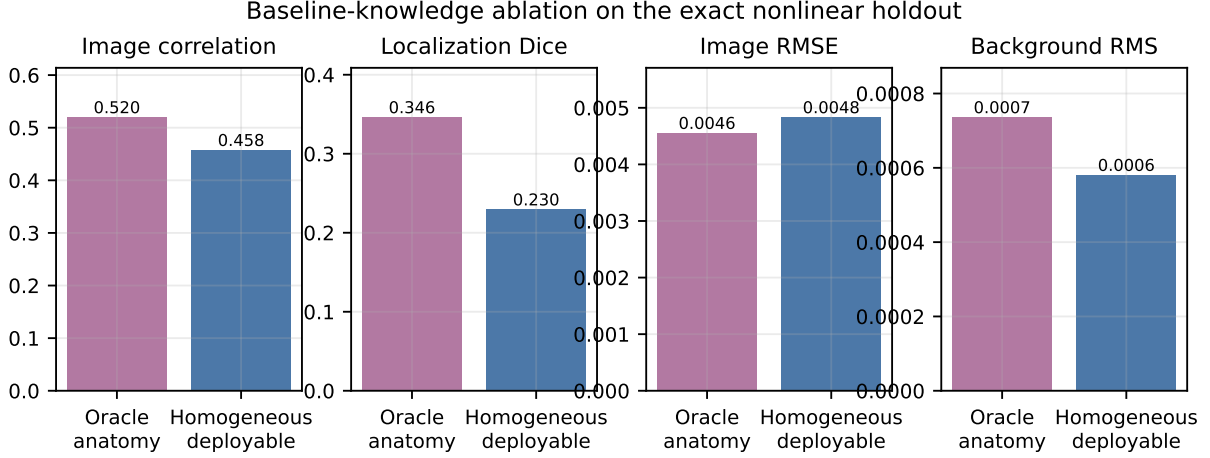


Figure 6: Explicit oracle-versus-deployable baseline comparison. The oracle result must not be quoted as real-data performance.

10.5 Architecture and loss ablations

Coordinates provide the largest isolated architectural improvement. Vessel weighting $\alpha = 1$ maximizes Dice and minimizes RMSE but also increases false background and voltage mismatch. A background coefficient of 0.25 recovers nearly the same correlation with lower background; larger coefficients continue suppressing background but erase vessel support. We therefore select the balanced $\alpha = 1, \beta = 0.25$ model rather than optimizing any single metric.

Table 10: Deployable homogeneous-baseline stage-2 ablations on 32 exact nonlinear holdout samples. “Mean corr.” is the mean of per-sample image correlations; “global corr.” pools all valid pixels.

Experiment	Global corr. $\uparrow$	Mean corr. $\uparrow$	RMSE $\downarrow$	Dice $\uparrow$	BG RMS $\downarrow$	Voltage RMS $\downarrow$
Direct core	0.458	0.419	0.00484	0.230	0.000581	2.73×10^{-6}
Proposal residual	0.390	0.331	0.00494	0.176	0.000474	1.36×10^{-6}
Direct + coordinates	0.561	0.545	0.00444	0.471	0.000865	1.73×10^{-6}
$\alpha = 1$	0.557	0.581	0.00430	0.543	0.001541	4.62×10^{-6}
$\alpha = 2$	0.524	0.507	0.00445	0.457	0.001778	5.93×10^{-6}
$\alpha = 4$	0.489	0.468	0.00477	0.407	0.002232	7.32×10^{-6}
$\alpha = 8$	0.479	0.454	0.00519	0.394	0.002730	1.02×10^{-5}
$\alpha = 1, \beta = 0.25$ (selected)	0.557	0.567	0.00432	0.485	0.001276	3.18×10^{-6}
$\alpha = 1, \beta = 0.5$	0.530	0.492	0.00446	0.462	0.000992	1.87×10^{-6}
$\alpha = 1, \beta = 1.0$	0.527	0.469	0.00453	0.427	0.000845	1.58×10^{-6}

Deployable homogeneous-baseline ablation sequence (* and hatching mark the selected experiment)

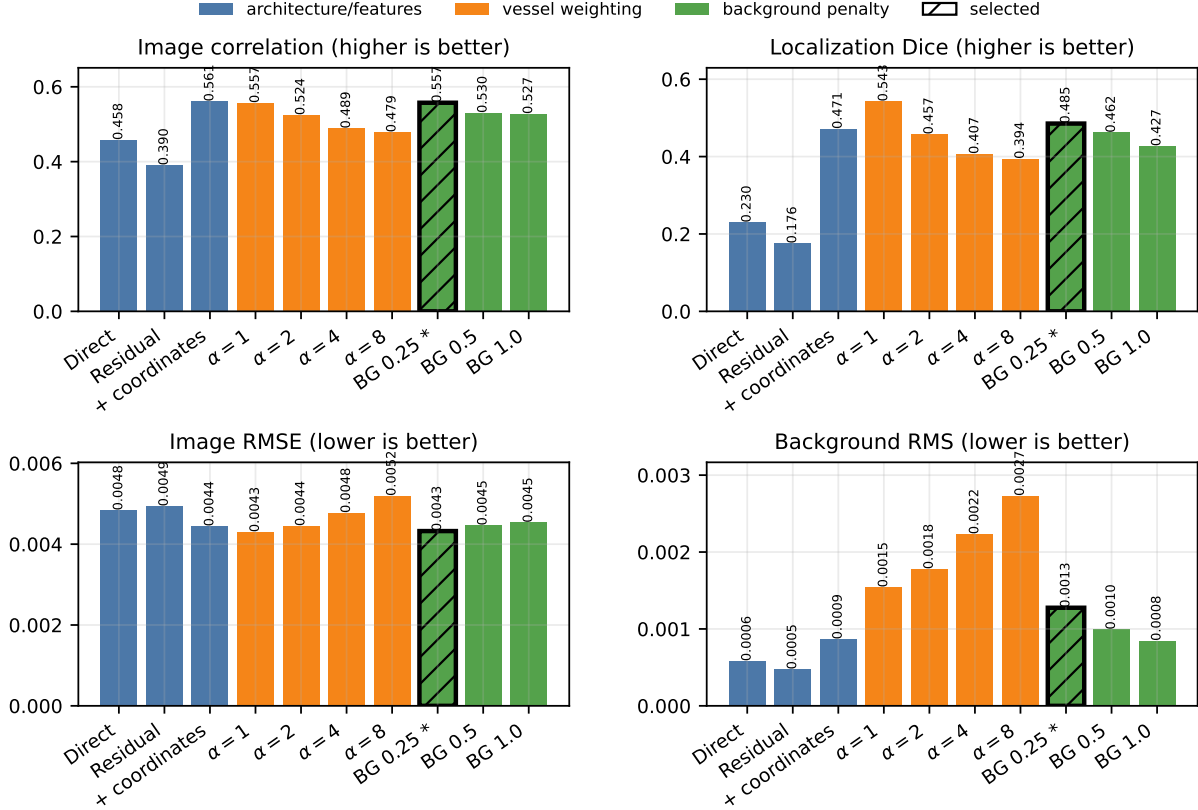


Figure 7: Complete deployable ablation sequence. The selected background-0.25 model is marked in the generated figure.

11 Qualitative output comparisons

Numerical averages answer whether an estimator improves on a population, but they do not show *how* it fails. This section therefore compares conductivity maps with fixed spatial registration and matched color scales. It separates three questions that must not be conflated:

- (i) did the earlier fixed-feature clean-truth experiment improve over its smoke run;
- (ii) how close were those earlier predictions to known synthetic ground truth; and
- (iii) how do the earlier outputs compare with real PVI reconstructions, for which anatomical conductivity ground truth is unavailable?

11.1 Smoke versus earlier fixed-feature experiment on an identical phantom

The anatomical smoke evaluation and sample 0 of the earlier exact nonlinear holdout contain the *same* clean conductivity target and the same Newton input to floating-point precision. This permits a direct experiment-evolution comparison. The previous model used one stage with only three training and one validation sample; the fixed-feature model used two stages with 512 training and 128 validation samples.

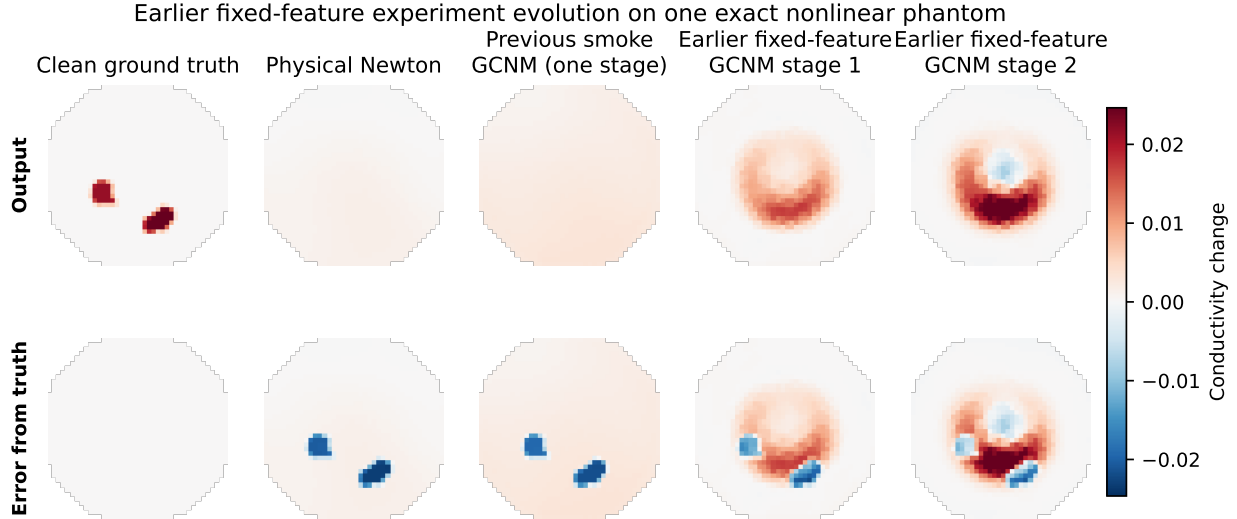


Figure 8: Direct output evolution on the same exact nonlinear phantom. The top row shows reconstructed conductivity change; the bottom row is prediction minus clean ground truth. The previous smoke GCN is nearly a spatially uniform bias. The fixed-feature models recover the broad vessel-bearing region, but not the two compact vessel boundaries; stage 2 additionally creates a central negative artifact.

Table 11: Pixel-matched metrics for the single shared phantom in [figure 8](#). This controlled example shows algorithm evolution, not population performance; the 32-sample averages remain in [table 7](#).

Method	Image corr. $\uparrow$	Element RMSE $\downarrow$	BG-RMS $\downarrow$	Dice $\uparrow$
Newton	0.258	0.00296	0.00075	0.455
Previous smoke GCN	0.182	0.00343	0.00215	0.000
Fixed-feature GCN stage 1	0.317	0.00323	0.00247	0.091
Fixed-feature GCN stage 2	0.319	0.00430	0.00400	0.136

The fixed-feature method is clearly more structured than the smoke model, but this example also demonstrates why image appearance or correlation alone is insufficient. The learned stages improve image correlation while worsening RMSE, background energy, and Dice relative to Newton on this particular phantom. The diffuse arc has roughly the correct global location but not the correct vessel support.

11.2 Ground truth versus earlier fixed-feature predictions

To avoid selecting only a visually favorable example, [figure 9](#) selects the best, median, and worst stage-2 cases by image correlation among all 32 exact nonlinear samples. The selected sample indices are 30, 26, and 21 with correlations 0.641, 0.423, and 0.147, respectively. Each row uses one shared conductivity scale, so amplitude and background errors remain visible.

Earlier fixed-feature exact-nonlinear holdout: objective case selection

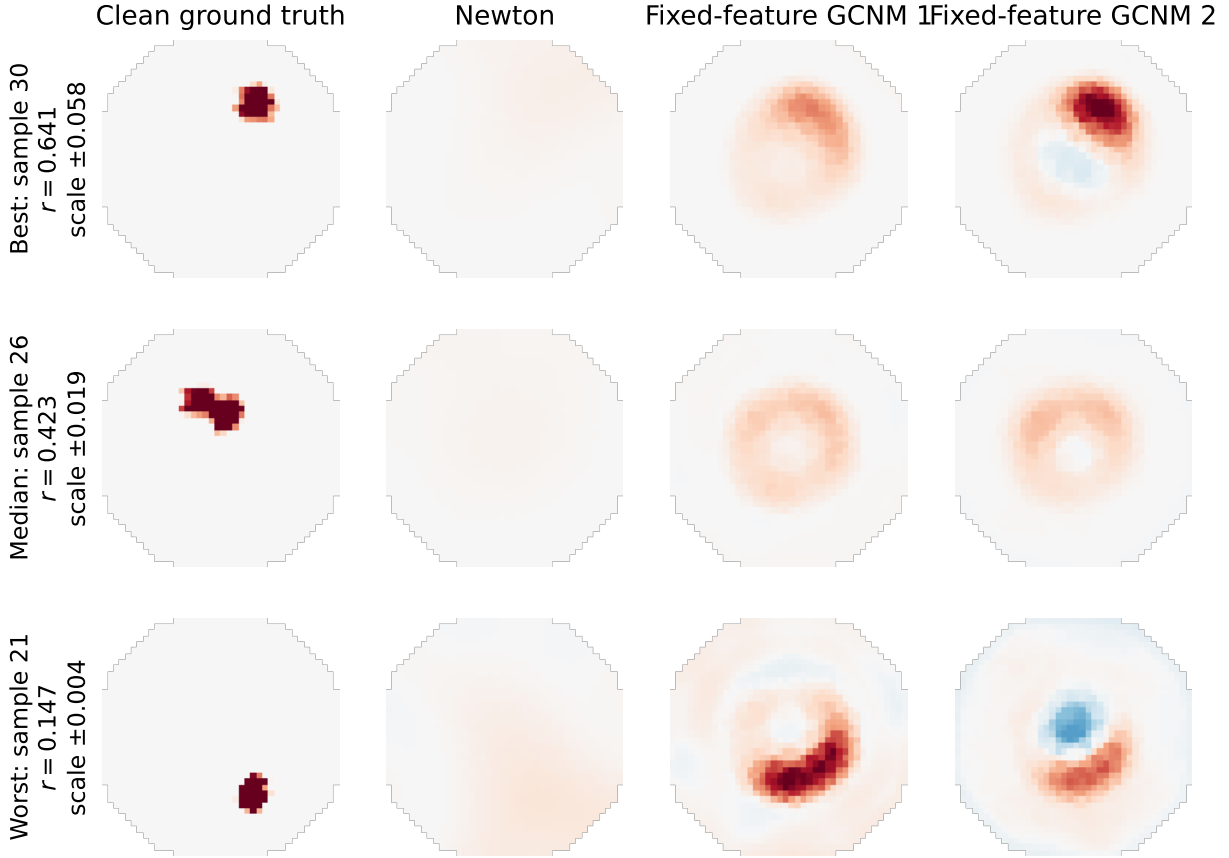


Figure 9: Clean ground truth, physical Newton reconstruction, and the two earlier fixed-feature GCN stages for objectively selected nonlinear-holdout cases. GCNM usually moves energy toward the correct region, but predicts a smooth distribution learned from the anatomical prior rather than resolving the compact ground-truth vessels. The worst case also shows sign-changing background structure.

11.3 Selected faithful model versus ground truth and prior outputs

The selected faithful gallery repeats the best/median/worst protocol using final stage image correlation over all 32 nonlinear holdout samples. Every row includes the clean target, historical saved Newton, iterative LM control, earlier fixed feature GCN, and both faithful stages. This is the most direct visual answer to whether recomputed physics and the controlled ablations improved the intermediate conductivity representation.

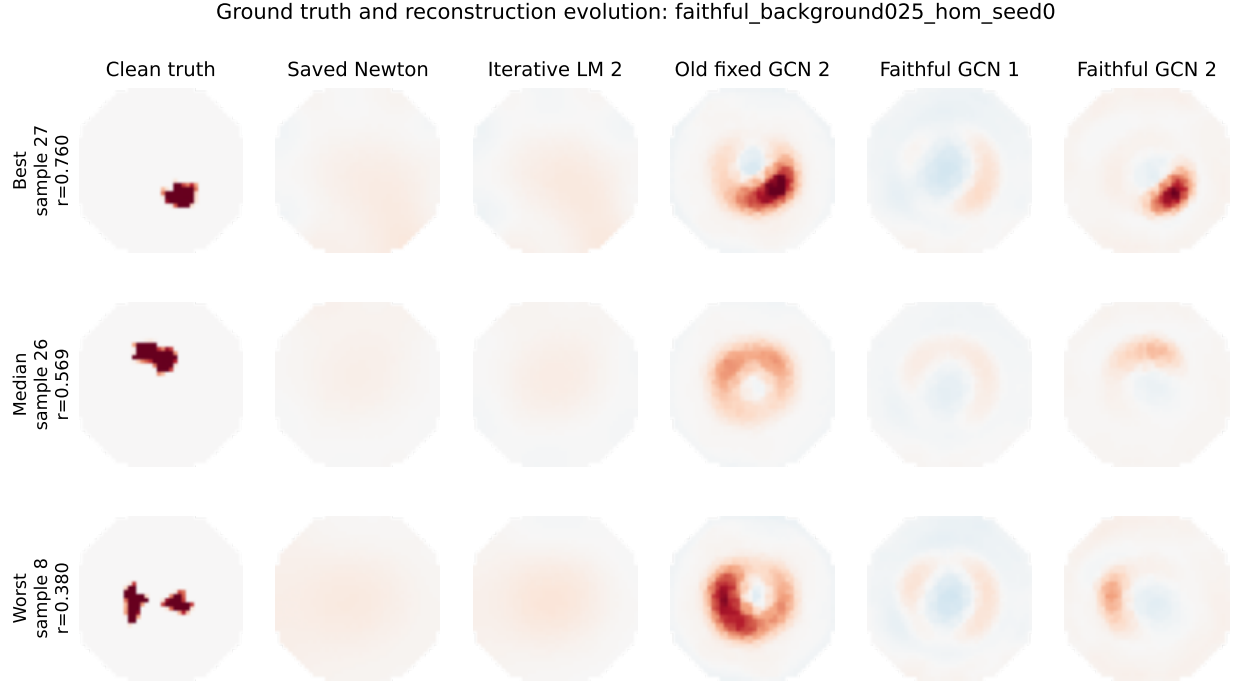


Figure 10: Clean ground truth and reconstruction evolution for the selected $\alpha = 1, \beta = 0.25$ homogeneous-baseline model. Rows are selected objectively as best, median, and worst final-stage image correlation; the color scale is shared within each row.

11.4 Comparison with archived PVI images

The real PVI images have no paired anatomical conductivity truth. They are themselves regularized one-step Newton reconstructions produced by the MATLAB pipeline. Thus, the only valid pixelwise question is whether our production port and earlier GCN reproduce those images—not whether either image is anatomically correct.

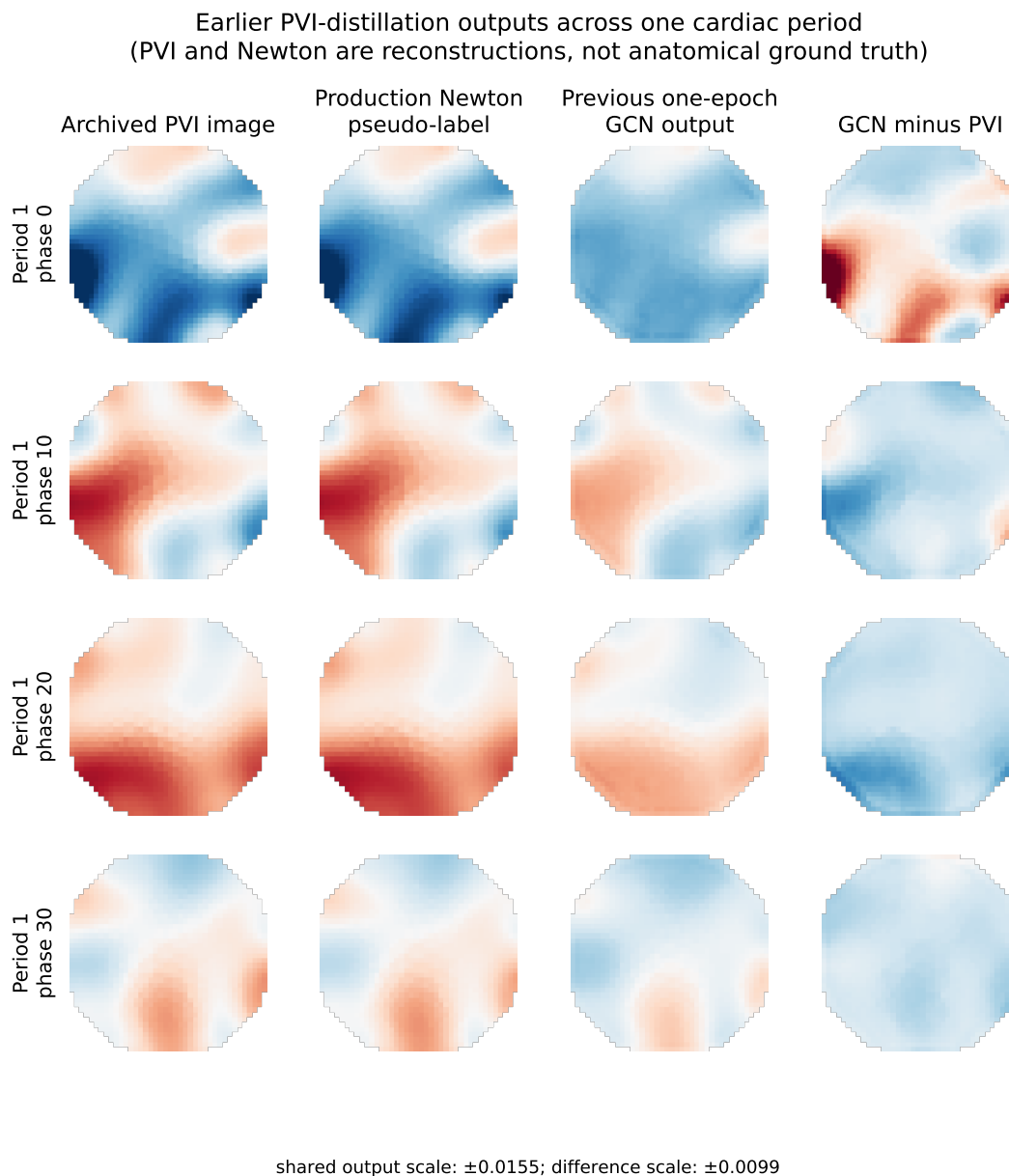


Figure 11: Four phases from subject 6, period 1. Columns compare the archived MATLAB PVI image, our production Newton pseudo-label, and the earlier one-epoch GCN output; the final column is GCN minus archived PVI. The first two columns agree closely, whereas the GCN attenuates and smooths the image. These displayed examples belong to the ten-sample smoke train/validation pack and are descriptive, not independent test evidence.

Table 12: Meaning of the real-PVI comparisons. Correlation with PVI measures reproduction of the existing inverse, not anatomical recovery.

Comparison	Samples	Image corr.	Interpretation
Production pseudo-label vs. archived PVI	10	0.9929	MATLAB/Python inverse parity
Previous GCN vs. archived PVI	10	0.9628	Smoke train/validation imitation
Previous GCN vs. production pseudo-label	10	0.9715	Distillation fit, not clean truth
Previous GCN vs. pseudo-label, held-out trials	4	0.9828	Independent trials 10–11, but the label is still PVI Newton

The final faithful model was also evaluated on all 1,600 test phases from trials 10–11. The iterative-LM panels below are reused from the shared homogeneous-core evaluation, while the faithful panels are produced by the selected background-0.25 checkpoint. The reference is deliberately labeled “pseudo-reference”: agreement or disagreement quantifies relation to the existing PVI inverse, never anatomical correctness.

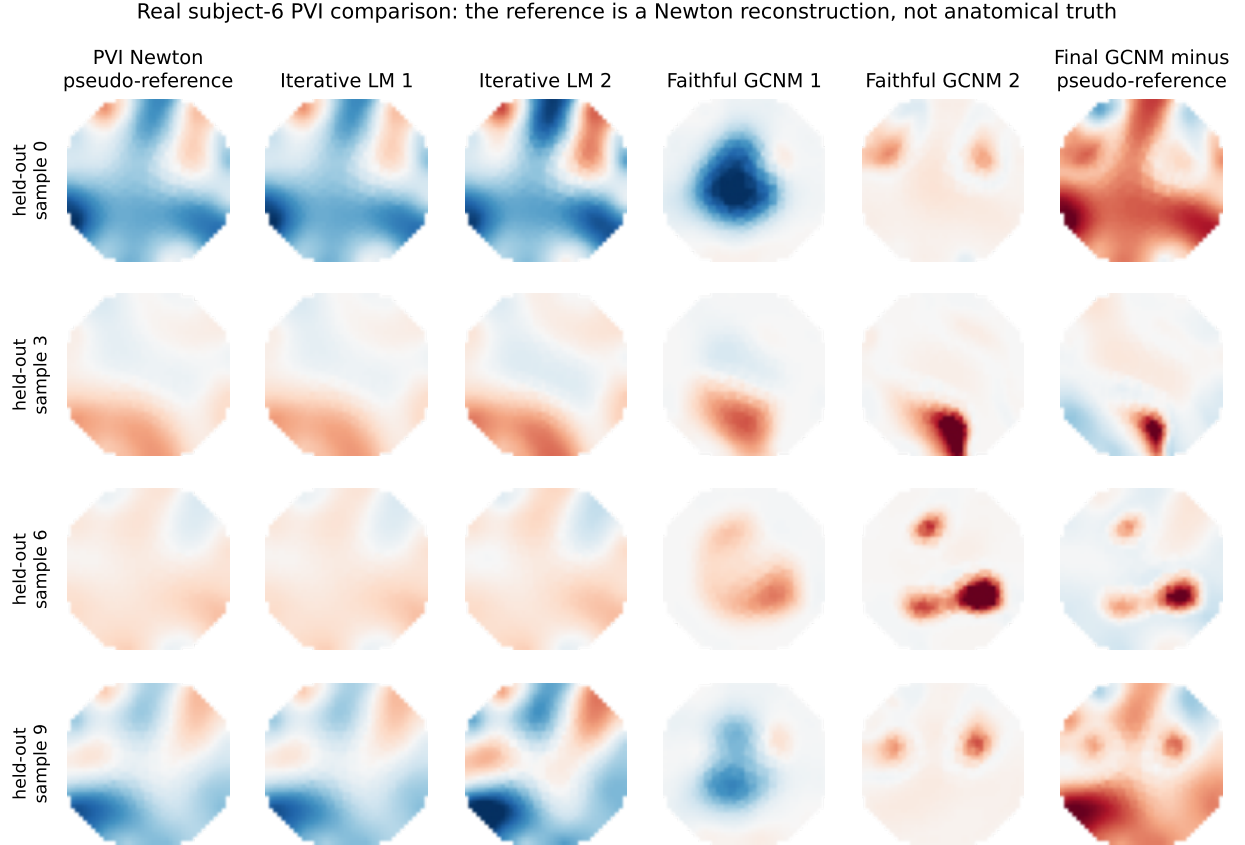


Figure 12: Full-test real subject-6 comparison for deterministic held-out phases. The PVI Newton pseudo-reference, two iterative-LM stages, two selected faithful GCNM stages, and final difference map are registered on the same inverse mesh. No synthetic Dice or vessel-ground-truth claim is transferred to this panel.

Table 13: Full 1,600-phase real-PVI pseudo-reference comparison. Correlation and RMSE are relative to the production Newton image, not anatomical truth. The first LM step reproduces that reference by construction.

Method	Global image corr.	Mean per-sample corr.	Image RMSE	Post-stage voltage RMS
Iterative LM stage 1	1.000	1.000	2.42×10^{-10}	6.66×10^{-6}
Iterative LM stage 2	0.977	0.971	0.00279	5.98×10^{-6}
Faithful direct stage 1	0.619	0.724	0.00821	2.00×10^{-5}
Faithful direct stage 2	0.212	0.348	0.00921	2.06×10^{-5}
Faithful proposal-residual stage 1	0.821	0.927	0.00686	1.79×10^{-5}
Faithful proposal-residual stage 2	0.424	0.555	0.00826	2.00×10^{-5}
Selected $\alpha = 1, \beta = 0.25$ stage 1	0.604	0.443	0.00713	1.68×10^{-5}
Selected $\alpha = 1, \beta = 0.25$ stage 2	0.169	0.275	0.01059	2.63×10^{-5}

The selected model reduces the physical voltage residual at stage 1 but stage 2 reverses that gain and departs further from the production inverse. This is direct evidence of synthetic-to-real distribution shift and a reason to expose intermediate stages rather than only the final image. It does not identify which image is more anatomically correct; that decision requires an external reference.

Table 14: Why the clean-truth and PVI panels must be read differently.

Panel	Target/reference	What can be concluded	What cannot be concluded
Earlier and faithful anatomical	Known fine-mesh vessel conductivity	Localization and amplitude error against truth	Immediate visual realism on a human acquisition
Archived PVI	MATLAB Newton reconstruction	Port parity and temporal consistency	Anatomical correctness or vessel resolution
Previous PVI GCN	Same Newton image used as feature and label	Ability to imitate/compress the existing inverse	Improvement beyond Newton
Cross-domain visual comparison	Unpaired synthetic and real samples	Gross scale, smoothness, and artifact differences	Pixelwise accuracy or subject-specific anatomy

12 Interpretation at bachelor and doctoral levels

12.1 Why correlation improves while RMSE worsens

Correlation is invariant to positive global scaling and rewards matching spatial variation. RMSE penalizes amplitude and background at every element. A prediction can therefore place high values near the correct vessel but remain too broad or too large:

$$\hat{\sigma} = a\sigma^* + b_{\text{diffuse}}, \quad a > 1, \quad (36)$$

yielding high correlation and poor RMSE. The vessel-weighted objective in [equation \(31\)](#) explicitly makes this outcome more likely because missing a vessel costs more than creating moderate background.

12.2 Why the earlier stage 2 amplified background

In the original algorithm, stage 2 is constrained by a new residual and Jacobian. In the earlier fixed-feature experiment, both stages saw the same Newton feature n . Stage 2 could use the stage-1 image as context, but no new physical discrepancy identified unsupported features. Reapplying the learned prior therefore sharpened localization while amplifying artifacts. The faithful method fixes this methodological defect by recomputing $\mathbf{r}_k$, $\mathbf{J}_k$, and $\mathbf{p}_k$. It does not, however, guarantee monotone voltage descent: a learned direct output can still trade data consistency for a conductivity prior, which is why voltage RMS remains an explicit reported metric.

12.3 Identifiability and learned hallucination

Let the singular value decomposition be $J = U\Sigma V^\top$. Only right singular vectors with nonzero singular values are observed. A learned network supplies components in the approximate nullspace:

$$\hat{\sigma} = J^\dagger \Delta \mathbf{v} + (I - J^\dagger J) \mathcal{P}_\theta(\Delta \mathbf{v}), \quad (37)$$

where $\mathcal{P}_\theta$ is a learned prior. The second term can improve a target drawn from the same distribution, but it can also hallucinate under distribution shift. Equation (37) is a useful conceptual decomposition, not the literal network implementation.

12.4 Why the original problem is easier

The original protocol supplies 928 measurements, whereas PVI supplies 32. Even before considering noise and electrode mismatch, the PVI Fisher information $J^\top \Sigma_\epsilon^{-1} J$ has rank at most 32. The original method also trains and tests on generic ellipses from nearly the same simulator, while our strongest test uses an exact nonlinear forward solve after training on a linearized Jacobian bank. Direct performance expectations must account for both differences.

13 Implemented faithful solver and execution safeguards

The faithful equations are implemented exactly as equations (32) and (33); no unimplemented whitening matrix or alternative normal equation is inserted in the report. The real PVI pack stores the MATLAB display-sign convention, so evaluation multiplies saved voltages by -1 before physical inversion and records that sign in its JSON report. Synthetic solver voltages use sign $+1$.

The deployable baseline is homogeneous 0.7 S/m in training and inference. The saved-anatomy mode is explicitly named oracle, and checkpoint loading rejects an incompatible baseline contract. Configuration, mesh, mapping, and dataset hashes make accidental cross-experiment reuse detectable. Positivity projection counts are reported at every stage; a zero count establishes that a run did not rely on clipping, although it does not prove physiological validity.

14 Executed ablations and remaining validation

14.1 Completed controlled sequence

The faithful core, iterative-LM control, direct/residual output comparison, coordinate ablation, vessel-weight sweep, background-penalty sweep, and composite checkpoint selection were all executed. The principal findings are:

1. recomputed LM physics alone improves Dice but cannot match learned spatial correlation;
2. direct output outperforms proposal-residual output on clean reconstruction;
3. coordinates materially improve the eight-electrode inverse because the mesh prior is strongly location dependent;
4. $\alpha = 1$ is preferable to larger vessel weights, which progressively amplify background and voltage inconsistency; and
5. $\beta = 0.25$ is the balanced selected model, while $\beta = 0.5$ and 1.0 demonstrate the expected noise-localization Pareto frontier.

14.2 Scale and uncertainty

Run at least five simulator seeds and report confidence intervals. Increase the number of independent anatomical Jacobian backgrounds. Evaluate calibration by ensembles or Monte Carlo dropout, and test whether uncertainty increases in low-sensitivity/nullspace regions.

14.3 Real-data validation hierarchy

Recommended evidence, from strongest to weakest, is:

1. physical tank/limb phantom with known inclusion location and conductivity;
2. co-registered ultrasound or another independent vessel-location target;
3. held-out forward-voltage consistency with realistic electrode nuisance;
4. temporal repeatability and cardiac-phase coherence;
5. downstream blood-pressure prediction;
6. visual comparison with PVI images.

The last item alone cannot establish anatomical validity because PVI is itself a regularized reconstruction.

15 Reproducibility map

Table 15: Public implementation and reproducibility files.

Path	Role
<code>gcnm_pvi/gcnm_physics.py</code>	PVI forward/Jacobian and generic iterative LM interface.
<code>gcnm_pvi/iterative_physics.py</code>	Stage-dependent differential residual, Jacobian, LM proposal, and diagnostics.
<code>gcnm_pvi/gcnm_model.py</code>	Direct and proposal-residual graph blocks.
<code>gcnm_pvi/anatomical_phantoms.py</code>	Shared coarse/fine anatomical generator.
<code>gcnm_pvi/generate_anatomical_dataset.py</code>	Linearized-bank and exact nonlinear datasets.
<code>gcnm_pvi/train_faithful_gcnm.py</code>	Greedy faithful training, losses, checkpoint contract, and hashes.
<code>gcnm_pvi/evaluate_faithful_gcnm.py</code>	Newton, LM, faithful GCNM, voltage, and real-PVI evaluation.
<code>gcnm_pvi/summarize_faithful_results.py</code>	Cross-experiment metric summaries.
<code>configs/subject006_anatomical_gcnm.yaml</code>	Full clean experiment configuration.
<code>scripts/submit_faithful_experiment.sh</code>	Dependency-controlled training and evaluation submission.
<code>scripts/submit_faithful_weight_sweep.sh</code>	Coordinate and vessel-weight ablations.
<code>scripts/submit_faithful_background_sweep.sh</code>	Background-penalty ablations.
<code>slurm/launch_train_faithful.sh</code>	Fixed-resource cluster training endpoint.
<code>slurm/launch_evaluate_faithful.sh</code>	Fixed-resource cluster evaluation endpoint.

Path	Role
tests/test_iterative_physics.py	Physics sign, descent, and per-stage recomputation tests.
tests/test_faithful_model.py	Proposal-residual initialization test.
reports/gcnm_pvi_latex/make_faithful_figures.py	Static figures from private result artifacts.

16 Conclusions

Injun Lee’s original code is a greedy unrolled inverse solver: nonlinear physics is recomputed, an LM update is formed, and a stage-specific GCN directly predicts the next conductivity. Our PVI work successfully replaced PyEIT with the PVI CEM solver, resolved subject-6 US120 geometry, reproduced MATLAB production output, created clean anatomical vascular supervision, introduced anti-inverse-crime forward/inverse discretizations, and evaluated on an exact nonlinear holdout.

The project now contains and has executed the faithful nonlinear mechanism, not only a proposal. On the exact nonlinear holdout, two LM-only iterations reach image correlation 0.291 and Dice 0.394. The selected faithful homogeneous model reaches correlation 0.557, RMSE 0.00432, Dice 0.485, and background RMS 0.00128. It improves structural recovery substantially over saved Newton and the earlier fixed-feature experiment while exposing a measurable trade-off between vessel support, background suppression, and voltage consistency.

The correct conclusion is still not that eight-electrode GCNM has recovered true human vascular anatomy. The inverse has 32 measurements for 1,330 element unknowns, and learned priors necessarily act in poorly observed subspaces. The completed real-PVI evaluation establishes execution, sign, temporal registration, and pseudo-reference comparison only. The next scientific evidence must come from multiple simulator seeds and uncertainty intervals, more independent anatomical backgrounds, a physical phantom, and co-registered ultrasound or another external anatomical reference.

A Pseudocode: original GCNM

```

Generate fine-mesh voltages V_s and coarse-mesh truths sigma*_s
Build FEM element graph G
Initialize sigma_0,s = 1 for every sample s

for k = 0,...,9:
  for each sample s:
    U_k,s, J_k,s = forward_and_normalized_jacobian(sigma_k,s)
    residual = U_k,s - V_s
    p_k,s = -(J^T J + lambda I)^(-1) J^T residual
    H_k,s = [sigma_k,s, p_k,s]

  train a new GCN_k to minimize mean ||GCN_k(H_k,s,G)-sigma*_s||^2
  sigma_{k+1},s = GCN_k(H_k,s,G) for all samples
  save GCN_k

```

B Pseudocode: implemented faithful differential PVI-GCNM

```

Choose one baseline policy and bind it in every checkpoint
Generate measured differential voltage dV_s
Initialize dsigma_0,s = 0
Cache U_base,s = F(sigma_b,s)

for k = 0,...,K-1:
    for each sample s:
        U_k, J_k = forward_and_jacobian(sigma_b,s + dsigma_k,s)
        residual = (U_k - U_base) - dV_s
        p_k,s = solve(J^T J + h_pvi^2 Rrec + lambda_lm I,
                      -J^T residual)
        H_k,s = [dsigma_k,s, p_k,s]

    train a separate GCN_k against clean dsigma*_s
    direct mode: dsigma_k+1,s = GCN_k(H_k,s,G)
    residual mode: dsigma_k+1,s = dsigma_k,s + p_k,s
                    + correction_k(H_k,s,G)
    project sigma_b,s + dsigma_k+1,s above conductivity floor
    save GCN_k plus baseline, physics, configuration, and hashes

```

C Metric definitions

For S samples and K elements,

$$\text{RMSE} = \sqrt{\frac{1}{SK} \sum_{s,i} (\hat{\sigma}_{s,i} - \sigma_{s,i}^*)^2}, \quad (38)$$

$$\text{BG-RMS} = \sqrt{\frac{\sum_{s,i} (1 - m_{s,i}) \hat{\sigma}_{s,i}^2}{\sum_{s,i} (1 - m_{s,i})}}, \quad (39)$$

$$\text{Dice}_{\text{volume}} = \frac{2|\hat{\mathcal{S}} \cap \mathcal{S}^*|}{|\hat{\mathcal{S}}| + |\mathcal{S}^*|}. \quad (40)$$

Here $\mathcal{S}^*$ is the true nonzero support and $\hat{\mathcal{S}}$ contains the same number of elements with largest predicted magnitude. Because the two sets have equal size, the implemented Dice reduces numerically to $|\hat{\mathcal{S}} \cap \mathcal{S}^*|/|\mathcal{S}^*|$.

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